" Code is like humor. When you have to explain it, it's bad. "

CSS Notes

Cascading Style Sheets

What is CSS ?

CSS (Cascading Style Sheets) is a style sheet language used to describe the presentation of a document written in HTML.

FEATURES

- ★ Easy to Learn
- ★ Easy to Use
- ★ Saves Time
- ★ Improves Website Look
- ★ Maintains Consistency
- ★ Powerful Styling Tool

By : Bikki Kumar Pandit

1. Introduction to CSS

What is CSS ?

→ CSS stands for Cascading Style Sheets.

→ It is a style sheet language used to describe the presentation of a document written in HTML.

→ CSS describes how HTML elements are to be displayed on screen, paper, or in other media.

CSS is used to :

- Style and design web pages
- Control layout and colors
- Change fonts, spacing, borders
- Make websites responsive and attractive

Why CSS ?

→ Separates content from presentation (HTML for structure, CSS for design)

→ Makes web pages more attractive and user-friendly.

→ Saves time and increases consistency.

→ Easy to maintain.

How CSS Works ?

HTML (Structure) Example : </> + CSS (Style) Example : { } → Result (Web Page) Example :

Key Points

- CSS works with HTML.
- CSS is case sensitive.
- CSS file has .css extension.
- CSS rules are read from top to bottom.

Remember

HTML gives structure
CSS gives style
Together they build
a beautiful website.

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2. CSS Syntax

CSS rule-set consists of a selector and a declaration block.

Selector ↓   Declaration Block

```
h1  {  color:  blue;
       font-size:  24px;  }
```

↑ Property ↑ Value ↑ Property ↑ Value

Parts of a Rule-set

1. Selector – selects the HTML element.
2. Declaration Block – contains one or more declarations.
3. Property – style feature you want to change.
4. Value – value of the property.

Example :

```
p {
    color: green;
    font-size: 18px;
}
```

→ This is a selector.

→ These are declarations.

→ Every declaration ends with a semicolon (;).

Notes

- CSS is not case sensitive (color and COLOR are same).
- A CSS rule-set must start with a selector.
- Declaration block is enclosed in curly braces { }.
- Each declaration has a property and a value.
- Each property and value pair is separated by a colon (:).
- Each declaration is ended with a semicolon (;).

3. Ways to Add CSS

There are three ways to add CSS in HTML.

1 Inline CSS

- → CSS is applied directly inside an HTML element using the style attribute.
- → It affects only that particular element.

Example :

```
<h1 style="color: red;
    font-size: 24px;">
    Hello CSS
</h1>
```

Result :

→ Hello CSS

2 Internal CSS

- → CSS is written inside the <style> tag in the <head> section.
- → It affects all elements in that HTML page.

Example :

```
<head>
  <style>
    h1 {
        color: blue;
        font-size: 24px;
    }
  </style>
</head>
<body>
  <h1>Hello CSS</h1>
</body>
```

Result :

→ Hello CSS

3 External CSS

- → CSS is written in a separate .css file and linked to the HTML file using the <link> tag.
- → It can be reused across multiple pages.

index.html

```
<head>
  <link rel="stylesheet"
     href="style.css">
</head>
<body>
  <h1>Hello CSS</h1>
</body>
```

+

style.css

```
h1 {
   color: purple;
   font-size: 24px;
}
```

Result in Browser : Hello CSS

Best Practice : External CSS is preferred because it keeps HTML clean, makes code reusable and easy to maintain.

4. CSS Selectors

CSS selectors are used to "select" HTML elements based on their id, class, type, attribute, etc.

1. Type Selector

→ Selects all elements of a specific type.

```css
p {
  color: blue;
  font-size: 18px;
}
```

Example :

HTML :

```html
<p>Welcome to CSS</p>
<p>This is a paragraph.</p>
```

CSS :

```css
p { color: blue; }
```

Result :

Welcome to CSS

This is a paragraph.

2. Class Selector

→ Selects elements with a specific class.

```css
.box {
  color: white;
  background-color: purple;
  padding: 10px;
}
```

Example :

HTML :

```html
<div class="box">Hello CSS</div>
```

CSS :

```css
.box { color: white;
       background-color: purple; }
```

Result : Hello CSS

3. ID Selector

→ Selects an element with a specific id.

```css
#main {
  color: darkgreen;
  font-weight: bold;
}
```

Example :

HTML :

```html
<h1 id="main">
  ID Selector
</h1>
```

CSS :

```css
#main { color: darkgreen;
  font-weight: bold; }
```

Result :

ID Selector

4. Universal Selector

→ Selects all elements in the document.

```css
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}
```

☆ It is useful for removing default margin and padding from all elements.

Note :

★ ID is unique in a page.

★ Class can be used multiple times.

Quick Tips

- Use class for groups of elements.

5. CSS Properties

{ CSS }

CSS properties are used to style HTML elements.
They control the layout, colors, fonts, spacing and more.

1 color

→ Sets the text color of an element.

Example :

```
h1 {
  color: blue;
}
```

→

Result :

Hello CSS

★ Changes the text color to blue.

2 font-size

→ Sets the size of the text.

Example :

```
p {
  font-size: 20px;
}
```

→

Result :

This is a paragraph

★ Increases the text size.

3 font-family

→ Sets the font style for text.

Example :

```
p {
  font-family: Arial, sans-serif;
}
```

→

Result :

This is a paragraph

★ Changes the font style.

4 background-color

→ Sets the background color of an element.

Example :

```
div {
  background-color: yellow;
}
```

→

Result :

This is a div

★ Changes the background color to yellow.

5 text-align

→ Aligns the text horizontally.

Example :

```
p {
  text-align: center;
}
```

→

Result :

This text is
center aligned

★ Centers the text.

Note :

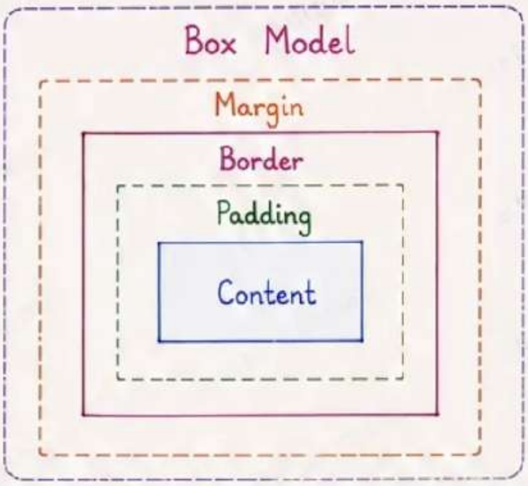
All properties are written

More Useful Properties

- border – sets border
- padding – space inside
- display – changes display
- text-decoration – underline, etc.

6. The CSS Box Model

Every HTML element is treated as a rectangular box in CSS. The box model consists of:
Content, Padding, Border, Margin.



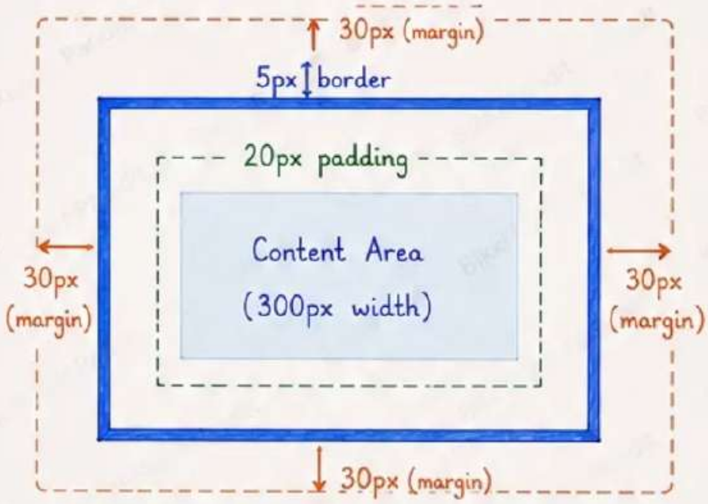
① Explanation

- Content – The actual content (text, image, etc.)
- Padding – Space between content and border.
- Border – A line around the padding (and content).
- Margin – Space outside the border.

② Example

```css
div {
   width: 300px;
   padding: 20px;
   border: 5px solid blue;
   margin: 30px;
   background-color: #e8f4ff;
}
```

→



③ Width & Height in Box Model

The total width of the box =
margin (L+R) + border (L+R) + padding (L+R) + content width

The total height of the box =
margin (T+B) + border (T+B) + padding (T+B) + content height

By default, width and height apply only to the content area.

④ Box-sizing Property

It controls how the total width and height of an element is calculated.

```css
/* Default */
  box-sizing: content-box;

/* Better */
  box-sizing: border-box;
```

content-box (default)	border-box (better)
Width and height are applied only to the content area.	Width and height include content, padding and border.
Padding and border are added to the total size.	Total size remains the same as given.

Example

width: 300px;
padding: 20px;
border: 5px;

content-box
total width = 300 + 40 + 10
= 350px

border-box

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7. CSS Display Property

The display property specifies how an element is displayed on the web page.

The most commonly used values are : block, inline, inline-block, none, flex

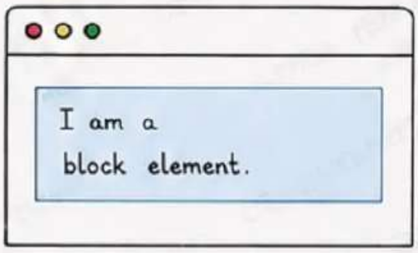
1 display: block

- Takes full width available.
- Starts on a new line.
- Height and width can be set.
- Example elements: div, h1 - h6, p, form, header, footer, etc.

Example :

```
<style>
  div {
      display: block;
      width: 300px;
      background-color: #d8e8ff;
      padding: 10px;
      margin: 10px 0;
  }
</style>
<div> I am a block element. </div>
```

Result in Browser :



★ Starts in a new line and takes full width.

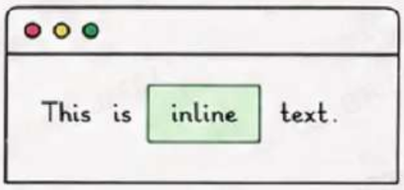
2 display: inline

- Does not start on a new line.
- Stakes only necessary width.
- Height and width cannot be set.
- Example elements: span, a, img, input, label, etc.

Example :

```
<style>
  span {
    display: inline;
    color: blue;
    background-color: #e0ffe0;
    padding: 4px 8px;
  }
</style>
<p>This is <span> inline </span> text.</p>
```

Result in Browser :



★ Does not start a new line and takes only content width.

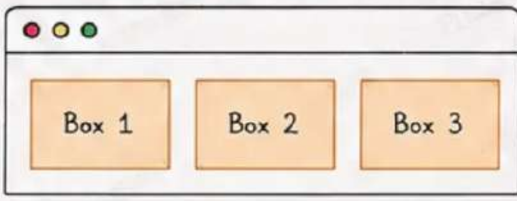
3 display: inline-block

- Does not start on a new line.
- Height and width can be set.
- Takes only necessary width.
- Useful for buttons, menus, etc.

Example :

```
<style>
  .box {
      display: inline-block;
      width: 100px;
      height: 60px;
      background-color: #ffe8d6;
      margin: 5px;
      text-align: center;
      line-height: 60px;
      border: 1px solid #ff9a55;
  }
</style>
<div class="box"> Box 1</div>
<div class="box"> Box 2</div>
<div class="box"> Box 3</div>
```

Result in Browser :



★ All boxes are in the same line and we can set width, height, margin, padding etc.

4 display: none

- Hides the element.
- Not visible on page.
- Does not take any space.

Example :

```
.hidden {
   display: none;
}
```

Useful when you want to show/hide element using JavaScript.

5 display: flex

- Used for flexbox layout.
- Makes it easy to align items in container.
- Powerful for modern

Example :

```
.container {
  display: flex;
  justify-content: center;
  align-items: center;
```

8. CSS Text & Font

{ CSS }

CSS helps us style the text and fonts on a webpage.
It makes our content look beautiful and easy to read.

1 Text Color

→ Sets the color of any text.

→ Example: color: red;

Example :

```
p {
   color: blue;
}
<p>This is a paragraph.</p>
```

Result :

This is a paragraph.

2 Text Alignment

→ Aligns text horizontally.

→ Values: left, center, right, justify

Example :

```
h1 {
   text-align: center;
}
<h1>Welcome to CSS</h1>
```

Result :

Welcome to CSS

3 Text Decoration

→ Adds decoration to text.

→ Values: underline, overline, line-through, none

Example :

```
a {
   text-decoration: underline;
}
<a href="#">Link</a>
```

Result :

Link

4 Text Transform

→ Changes the case of text.

→ Values: capitalize, uppercase, lowercase

Example :

```
p.upper { text-transform: uppercase; }
p.cap { text-transform: capitalize; }
p.low { text-transform: lowercase; }
<p class="upper">hello world</p>
<p class="cap">hello world</p>
<p class="low">HELLO WORLD</p>
```

Result :

HELLO WORLD
Hello World
hello world

5 Font Family

→ Sets the font type.

→ Example: font-family: Arial, sans-serif;

Example :

```
p { font-family: 'Times New Roman', serif; }
<p>This is in Times New Roman font.</p>
```

Result :

This is in Times New Roman font.

6 Font Size

→ Sets the size of text.

→ Example: font-size: 20px;

Example :

```
p { font-size: 20px; }
<p>This is a big text.</p>
```

Result :

This is a big text.

7 Font Style

→ Changes the style of text.

→ Values: normal, italic, oblique

→ Example :

→ { font-style: italic; }

<p>This is italic text.</p>

Result :

This is italic text.

Note :

Text properties make your ...

Quick Tips

8. CSS Position Property

The position property specifies the type of positioning method used for an element.

Types : static relative absolute fixed sticky

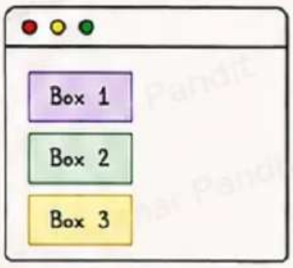
1 position: static (default)

- The element is positioned according to the normal flow of the page.
- Top, right, bottom, left and z-index have no effect.
- The element does not leave a gap in the page.

Example :

```css
.box {
    position: static;
    top: 20px; /* No effect */
    left: 20px; /* No effect */
}
```

Result in Browser :



★ Elements are placed in normal flow

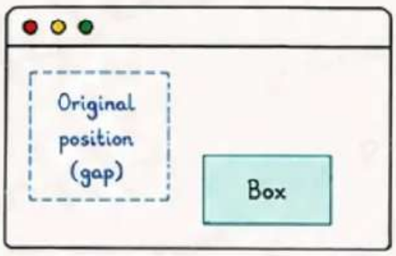
2 position: relative

- The element is positioned relative to its normal position.
- It can be moved using top, right, bottom, left.
- It leaves the gap at the original position.
- z-index works.

Example :

```css
.box {
    position: relative;
    top: 20px;
    left: 30px;
    background: #d1ecf1;
}
```

Result in Browser :



★ Moved fro original position b gap remai

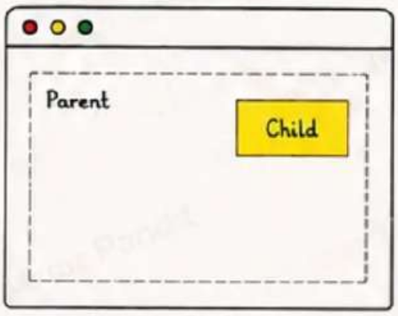
3 position: absolute

- The element is positioned relative to its nearest positioned ancestor (not static).
- If no positioned ancestor, it is relative to the initial containing block (page).
- It does not leave any gap.
- z-index works.

Example :

```css
.parent {
    position: relative;
    width: 300px;
    height: 150px;
    border: 2px dashed #999;
}
.child {
    position: absolute;
    top: 20px;
    right: 20px;
    background: #ffc107;
    padding: 5px 10px;
}
```

Result in Browser :



★ Placed rela to nearest positioned ancestor. No gap.

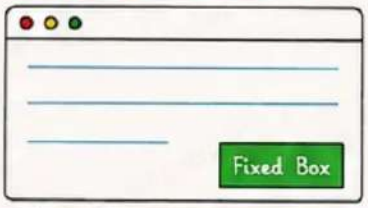
4 position: fixed

- The element is positioned relative to the viewport (browser window).
- It stays in the same place even when page is scrolled.
- It does not leave any gap.
- z-index works.

Example :

```css
.box {
    position: fixed;
    bottom: 20px;
    right: 20px;
    background: #28a745;
    color: white;
    padding: 8px 12px;
}
```

Result in Browser :



★ Remains fix at the sam place even on scroll.

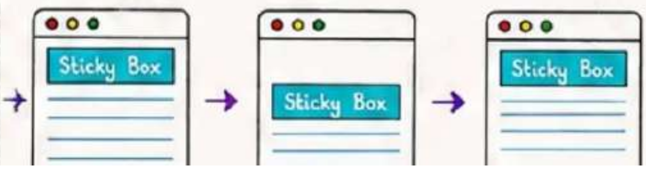
5 position: sticky

- A hybrid of relative and fixed.
- The element behaves like relative until a given scroll position.
- Then it sticks to a position like fixed.

Example :

```css
.box {
    position: sticky;
    top: 0;
    background: #17a2b8;
    color: white;
    padding: 10px;
}
```

Result in Browser :



★ Sticks top (bottom when reach

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9. CSS Position Property

{ CSS }

The position property specifies the type of positioning method used for an element.

Position values : static, relative, absolute, fixed, sticky

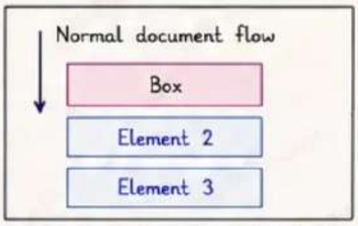
1 static (default)

- Default position of all elements.
- Follows normal document flow.
- top, right, bottom, left do not work.

Example :

```
.box {
   position: static;
}
<div class="box">Box</div>
```

Result in Browser :



★ Element is placed in normal flow.

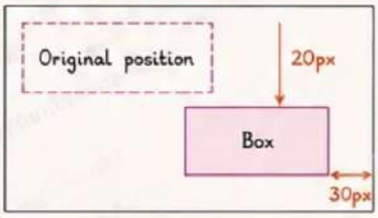
2 relative

- Position relative to its normal position.
- Leaves a gap at original position.
- top, right, bottom, left will work.

Example :

```
.box {
   position: relative;
   top: 20px;
   left: 30px;
}
<div class="box">Box</div>
```

Result in Browser :



★ Element moves from its original place but space is reserved.

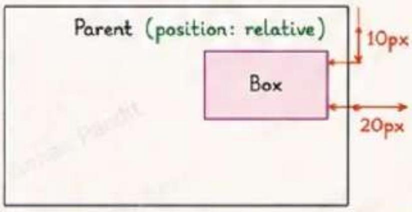
3 absolute

- Position relative to the nearest positioned ancestor (not static).
- Removed from normal flow.
- Can overlap other elements.

Example :

```
.parent { position: relative; }
.box {
   position: absolute;
   top: 10px;
   right: 20px;
}
```

Result in Browser :



★ Element is positioned inside parent and removed from flow.

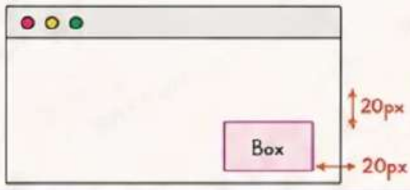
4 fixed

- Position relative to the viewport (browser window).
- Stays in the same place even when page scrolls.
- Removed from normal flow.

Example :

```
.box {
   position: fixed;
   bottom: 20px;
   right: 20px;
}
```

Result in Browser :



★ Element stays fixed in the viewport.

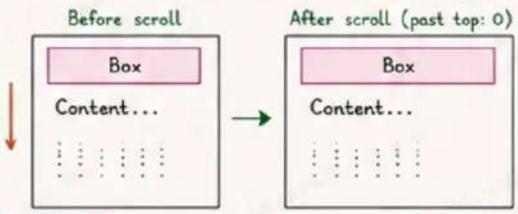
5 sticky

- Acts like relative until a scroll position is met.
- Then it becomes fixed.
- Useful for headers, menus.

Example :

```
.box {
   position: sticky;
   top: 0;
}
```

Result in Browser : (on scroll)



★ Sticks to the top after the scroll position is reached.

Note : Absolute and fixed elements are removed from the normal flow.

Quick Summary

Value	Position relative to	In normal flow?	Use Case
static	Normal document flow	Yes	Default positioning
relative	Its normal position	Yes (space reserved)	Small adjustments
absolute	Neerest positioned ancestor	No	Overlapping, inner elements
fixed	Viewport (browser window)	No	Header, buttons, chat icons
sticky	Scroll position	Yes (then fixed)	Sticky header, menus

Tip
Always set position: relative; on parent when using absolute for child.

10. CSS Background

The background properties are used to add background color, image, repeat, position, size and more to an element.

Common Properties : background-color, background-image, background-repeat, background-position, background-size, background-attachment, background.

1 background-color

- Sets the background color of an element.
- Values: any valid color name, hex, rgb, hsl, etc.

Example :

```css
.box {
  background-color: #d1e7ff;
  padding: 15px;
  border: 1px solid #3b82f6;
}
```

Result in Browser :

This box has a light blue background.

2 background-image

- Sets an image as the background of an element.
- Values: url('image.jpg')

Example :

```css
.box {
  background-image: url('nature.jpg');
  background-size: cover;
  height: 120px;
  border: 1px solid #555;
}
```

Result in Browser :

3 background-repeat

- Specifies how the background image repeats.
- Values: repeat, repeat-x, repeat-y, no-repeat

Example :

```css
.box {
  background-image: url('pattern.png');
  background-repeat: repeat-x;
  height: 80px;
  border: 1px solid #555;
}
```

Result in Browser :

↑ Image repeats only horizontally.

4 background-position

- Sets the starting position of the background image.
- Values: left, right, center, top, bottom, or pixel values.

Example :

```css
.box {
  background-image: url('nature.jpg');
  background-position: center;
  height: 120px;
  border: 1px solid #555;
}
```

Result in Browser :

5 background-size

- Sets the size of the background image.
- Values: auto, cover, contain, or specific size (px, %, etc.)

Example :

```css
.box {
  background-image: url('nature.jpg');
  background-size: cover;
  height: 120px;
  border: 1px solid #555;
}
```

Result in Browser :

cover → fills the entire area (may crop image).

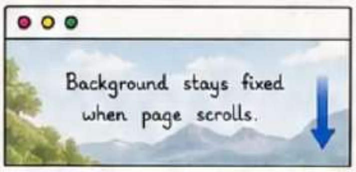
6 background-attachment

- Specifies if the background image scrolls with the page or stays fixed.
- Values: scroll, fixed, local

Example (fixed) :

```css
.box {
  background-image: url('nature.jpg');
  background-attachment: fixed;
  height: 200px;
}
```

Result in Browser :



7 background (Shorthand)

- Shorthand property to set all background properties in one line.
- Syntax: background: color image repeat position / size attachment;

Example :

```css
.box {
  background: #f0f8ff url('pattern.png')
        no-repeat center/cover fixed;
  height: 150px;
  border: 1px solid #555;
```

Result in Browser :

Shorthand example (all in one line)

Tip

Use background shorthand to

11. CSS Border

Borders are used to define the edge (boundary) of an element.

Common Properties : border, border-width, border-style, border-color, border-radius, border-top, border-right, border-bottom, border-left

1 border

- Shorthand property to set all border properties in one line.
- Syntax :
  border: width style color;

Example :

```
.box {
  border: 2px solid blue;
}
```

Result in Browser :

I have a border.

2 border-width

- Sets the width of the border.
- Values :
  thin, medium, thick, or length (px, em, etc.)

Example :

```
.box {
  border-width: 5px;
  border-style: solid;
  border-color: red;
}
```

Result in Browser :

Thick border (5px)

3 border-style

- Sets the style of the border.
- Values :

solid	————	double	════
dashed	- - - - -	groove	════
dotted	....	ridge	════

Example :

```
.box {
  border: 3px dashed
          green;
}
```

Result in Browser :

Dashed Border

4 border-color

- Sets the color of the border.
- Values :
  any valid color name, hex, rgb, etc.

Example :

```
.box {
  border: 2px solid
          #ff5722;
}
```

Result in Browser :

Orange Border

5 border-radius

- Rounds the corners of the border.
- Values :
  length (px, %, em) or per corner.

Example :

```
.box {
  border: 3px solid purple;
  border-radius: 15px;
}
```

Result in Browser :

Rounded Corners

Per Corner Radius
border-top-left-radius
border-top-right-radius
border-bottom-right-radius
border-bottom-left-radius

6 Individual Side Borders

- Set border for specific side.
- Properties :
  border-top
  border-right
  border-bottom
  border-left

Example :

```
.box {
  border-top: 4px solid blue;
  border-right: 2px dashed red;
  border-bottom: 3px dotted green;
  border-left: 5px double purple;
}
```

Result in Browser :

Different borders on each side

7 Border Example Table

Style	Example	Code
solid	————————	border: 2px solid black;
dashed	- - - - - - - - - - - -	border: 2px dashed blue;
dotted	....	border: 2px dotted green;
double	════════	border: 4px double purple;
groove	════════	border: 4px groove gray;
ridge	════════	border: 4px ridge brown;
none	No Border	border: none;

8 Border Shorthand Examples

Code	Result
border: 1px solid black;	
border: 3px dashed blue;	
border: 5px dotted green;	
border: 4px double purple;	
border: 2px solid #ff9800;	
border: thick groove gray;	

12. CSS Width, Height & Margin

Width and Height are used to set the size of an element.
Margin is used to create space outside the border of an element.

Common Properties : width, height, max-width, min-width, margin, margin-top, margin-right, margin-bottom, margin-left

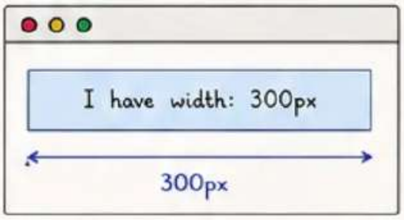
1 width

- Sets the width of an element.
- Values : length (px, %, em), auto

Example :

```css
.box {
  width: 300px;
  background: #d0e7ff;
  border: 1px solid #007bff;
  padding: 10px;
}
```

Result in Browser :



% is relative to parent element.

2 height

- Sets the height of an element.
- Values : length (px, %, em), auto

Example :

```css
.box {
  height: 150px;
  background: #e2ffd9;
  border: 1px solid #28a745;
}
```

Result in Browser :

Auto height adjusts according to content.

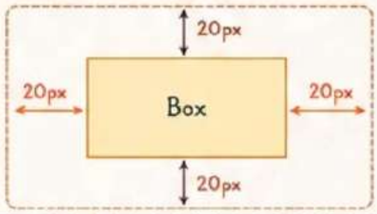
3 margin

- Creates space outside the border.
- Values : length (px, %, em), auto

Example :

```css
.box {
  width: 200px;
  margin: 20px;
  background: #fff3cd;
  border: 1px solid #ffc107;
}
```

Result in Browser :



margin: 20px; sets 20px margin on all 4 sides.

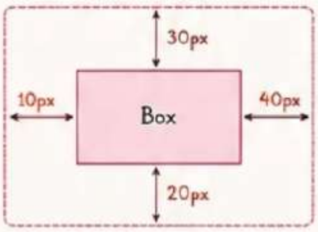
4 Individual Margin Properties

We can set margin for each side separately.

Example :

```css
.box {
  width: 200px;
  background: #f8d7da;
  border: 1px solid #dc3545;
  margin-top: 30px;
  margin-right: 40px;
  margin-bottom: 20px;
  margin-left: 10px;
}
```

Result in Browser :



Shorthand for Margin

Syntax	Meaning
margin: 20px;	Top Right Bottom Left = 20px
margin: 10px 20px;	Top & Bottom = 10px, Left & Right = 20px
margin: 10px 20px 30px;	Top = 10px, Left & Right = 20px, Bottom = 30px
margin: 10px 20px 30px 40px;	Top = 10px, Right = 20px, Bottom = 30px, Left = 40px

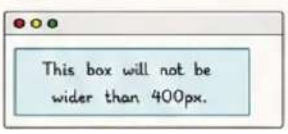
5 max-width & min-width

- Controls the maximum or minimum width.

Example (max-width)

```css
.box {
  max-width: 400px;
  background: #d1ecf1;
  padding: 10px;
  border: 1px solid #17a2b8;
}
```

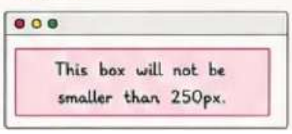
Result :



Example (min-width)

```css
.box {
  min-width: 250px;
  background: #fff0f6;
  padding: 10px;
  border: 1px solid #e83e8c;
}
```

Result :



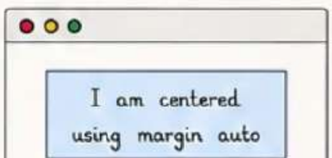
6 Auto Margin (Centering)

- Use auto margin to center block elements.

Example :

```css
.box {
  width: 300px;
  margin: 20px auto;
  background: #cce5ff;
  padding: 10px;
  border: 1px solid #004085;
}
```

Result in Browser :



margin-left: auto; and margin-right: auto; centers the element horizontally.

13. CSS Padding

Padding is used to create space inside the border of an element.

Common Properties : padding, padding-top, padding-right, padding-bottom, padding-left

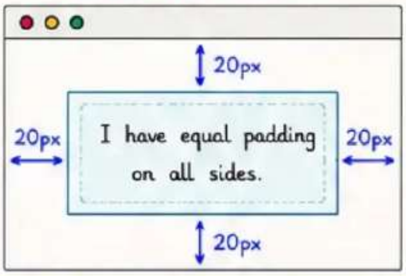
1 padding

- Creates equal padding on all four sides.
- Values : length (px, %, em), or auto

Example :

```css
.box {
    padding: 20px;
    background: #e0f7fa;
    border: 2px solid #00acc1;
}
```

Result in Browser :



Padding is inside the border. Content never touches the border.

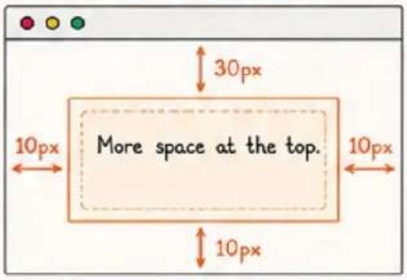
2 padding-top

- Adds padding only to the top.
- Values : length (px, %, em), or auto

Example :

```css
.box {
    padding-top: 30px;
    padding-right: 10px;
    padding-bottom: 10px;
    padding-left: 10px;
    background: #fff3e0;
    border: 2px solid #fb8c00;
}
```

Result in Browser :



Only top padding is 30px. Other sides remain 10px.

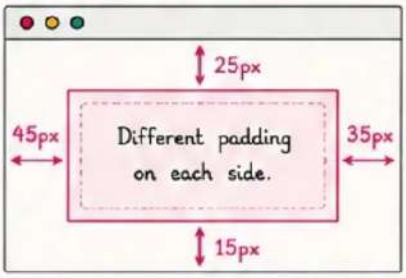
3 Individual Side Padding

- Set different padding for each side.
- Values: length (px, %, em), or auto

Example :

```css
.box {
    padding-top: 25px;
    padding-right: 35px;
    padding-bottom: 15px;
    padding-left: 45px;
    background: #fce4ec;
    border: 2px solid #e91e63;
}
```

Result in Browser :



top = 25px
right = 35px
bottom = 15px
left = 45px

4 Shorthand for Padding

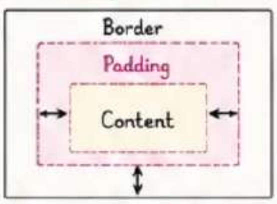
- We can set all paddings using shorthand.

padding: 20px;	all four sides
padding: 10px 20px;	top & bottom 10px, left & right 20px
padding: 10px 20px 30px;	top 10px, left & right 20px, bottom 30px
padding: 10px 20px 30px 40px;	top, bottom, left

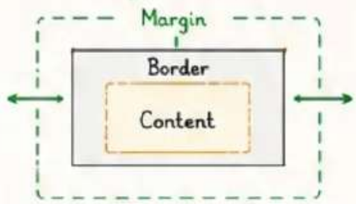
Example	Code	Visualization (Top | Right | Bottom | Left)
All sides equal	padding: 20px;	20px | 20px | 20px | 20px
Top & Bottom same	padding: 10px 20px;	10px | 20px | 10px | 20px
Top, LR, Bottom	padding: 10px 20px 30px;	10px | 20px | 30px | 20px
All four different	padding: 10px 20px 30px 40px;	10px | 20px | 30px | 40px

5 Padding vs Margin

Padding (inside)



Margin (outside)



Remember

- Padding increases the space inside an element.
- It pushes the content away from the border.
- It affects the size of the element.
- Padding has background color.

Bikki Kumar Pandit